Bedfordview Primary School
TERM 4 Year End Test

Name:	Grade 5 : A B C D
Subject: Natural Science and Technology	
Examiner: Mrs C. Nash	Date: 22 November 2024
Moderator: Mr J. Nash	Time Allocation: 1hour 30min Total: 60
INSTRUCTIONS 1. Read all the instructions and questions carefully. 2. Write neatly and legibly. 3. Write in full sentences unless otherwise instructed. 4. Look carefully at the mark allocation when giving your answers. 5. Answer in the spaces provided. This paper consists of 5 pages.	Parent's signature & comments: _____ _____ _____

1	2	3	4	5	6	7
0 – 29%	30 – 39%	40 – 49%	50 – 59%	60 – 69%	70 – 79%	80 – 100%

	60% Section A: Term 3 Energy & Electricity and Energy & Movement					40% Section B: Term 4 Planet Earth and Surface of the Earth				
Question	**1**	**2**	**3**	**4**	**5**	**1**	**2**	**3**	**Total**	**%**
Allocated mark	7	6	12	4	7	7	5	12	60	
Learner's mark										
Moderated mark										

Content: **Energy and Electricity**: cells and batteries; mains electricity. **Energy and Movement**: wheel and axle; elastics and springs. **Planet Earth**: the earth moves and **Surface of the Earth**
Skills: Understanding and interpreting data on storing of energy, circuits, electricity and transferring of energy. Identifying objects that are elastic. Understanding stored energy and movement of energy

SECTION A – Energy and Electricity & Energy and Movement /36

Question 1: Definitions (Knowledge recall) /7

Match the words in column A with their meaning in column B.

Column A	Column B
A. Components	1. Things you do to stop something dangerous happening.
B. Battery	2. Smallest whole part of a battery.
C. Precautions	3. Groups of related parts that work together as a whole for a particular purpose.
D. Systems	4. The parts that make up an electric circuit.
E. Cell	5. Kinetic energy.
F. Circuit	6. Group of cells.
G. Movement	7. A system that transfers electrical energy.

A-	B-	C-	D-	E-	F-	G-

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<u>Questions 2:</u> (True or False) **/6**

State whether the following statements are true or false.

2.1) Energy is stored in cells/batteries.

_____

2.2) An electrical circuit is a source of stored energy.

_____

2.3) A simple circuit consists of a cell and connecting wires.

_____

2.4) Mains electricity is electricity from the power station that is transferred by the national grid to homes and schools.

_____

2.5) A power station can use coal as a source of energy to produce electricity.

_____

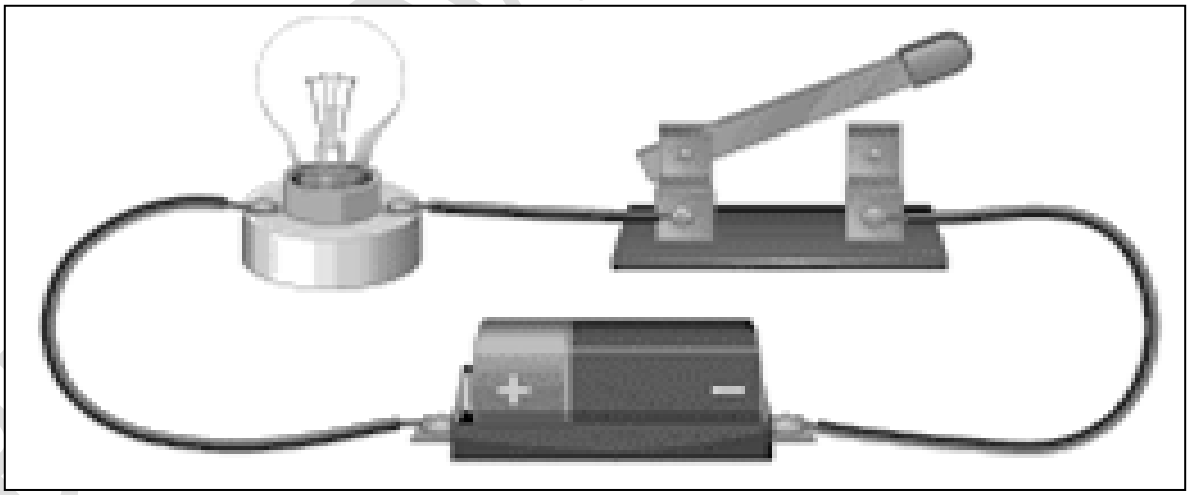
2.6) Safey precautions should not be taken into consideration when you work with electricity.

_____

<u>Question 3: Electrical Circuit</u>. **Total = 12**

Study the electrical circuit below and answer the questions that follow.

3.1 **Explain the function** of each component in the following electrical circuit: **/4**



Battery:_____

Switch:_____

Light bulb:_____

Copper wire:_____

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3.2 **Draw and label** a diagram of a closed simple circuit so that the globe will shine. **8 x ½ = /4**

- cell - light bulb - connecting wire - direction of energy flow

3.3 **Match** the word to the correct description. **/4**

Words	Description
1) Output Energy	a) a substance that can be eaten or burned to produce energy.
2) Fuels	b) saying what you think might happen.
3) Input Energy	c) energy such as heat and light that we get from burning fuel.
4) Predict	d) the amount of energy that must be added to fuel to make it burn.

1)_____ 2)_____ 3)_____ 4)_____

Question 4: True or False **/4**

State whether the following sentences are true or false.

a) The wheel is one of the oldest inventions. _____

b) Using a vehicle allows us to move around more easily. _____

c) The bulb in an electrical circuit will shines when it is connected to the cells by one wire.

_____

d) In a torch circuit, electrical energy is transferred from the light bulb to the cell.

_____

Question 5: Mains electricity /7

Place the following statements in their correct order:

How electricity flows from power stations to our homes, schools, and offices.	Order 1-7
1. Substations reduce high-voltage electricity to a lower voltage, which can be used in homes and offices.	
2. An electrical appliance draws the energy needed to work from the mains supply.	
3. Powerlines carry electricity from the power stations to substations in areas where people live and work.	
4. Wall sockets built into the walls of a building provide places for people to safely plug into the mains supply.	
5. When plugged into the wall socket, an appliance is connected to the mains electrical supply.	
6. Power stations burn fuels such as coal to generate electricity.	
7. Electricity boxes at home control the supply of electricity to the building.	

SECTION B – Planet Earth and Beyond / 24

Question 1: Definitions (Knowledge recall) /7

Match the words in column A with their meaning in column B.

Column A	Column B
A. Orbit	1. The hard layer on the outside of the Earth.
B. Rotation	2. Smal pieces of broken-down rock.
C. Conserve	3. The pathway that one object takes around another.
D. Crust	4. When dead plants and animals break down and become part of the soil.
E. Structure	5. To use wisely and to avoid wasteful or destructive use.
F. Decompose	6. The movement of an object around itself.
G. Sediments	7. How different parts are put together.

A-	B-	C-	D-	E-	F-	G-

Questions 2: (True or False) /5

State whether the following sentences are true or false.
If false, rewrite the statement to make it true.

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a) The Earth's axis is vertical. _____

_____

b) The whole Earth can receive light at the same time. _____

_____

c) The side of the Earth that receives light is experiencing night. _____

_____

d) The Sun takes 365 ¼ days to revolve around the Earth. _____

_____

e) The shape of the Earth's orbit is square. _____

_____

<u>Question 3:</u> Different layers of Earth's crust. **Total = /12**

3.1) **Complete** the following paragraph by filling in the missing words. **6 x ½ = /3**

_____ are big pieces of solid material. Rocks break down to make smaller rocks and _____ . Rocks form the _____ layer of land. The _____ and the _____ both sit on top of a layer of _____ .

3.2) **Draw and label** a diagram showing the <u>different layers of Earth's crust</u>. **/6**

<u>Diagram</u>	<u>Labels</u>

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3.3 **Complete** the following table by filling in the missing information. **3 x ½ = /1½**

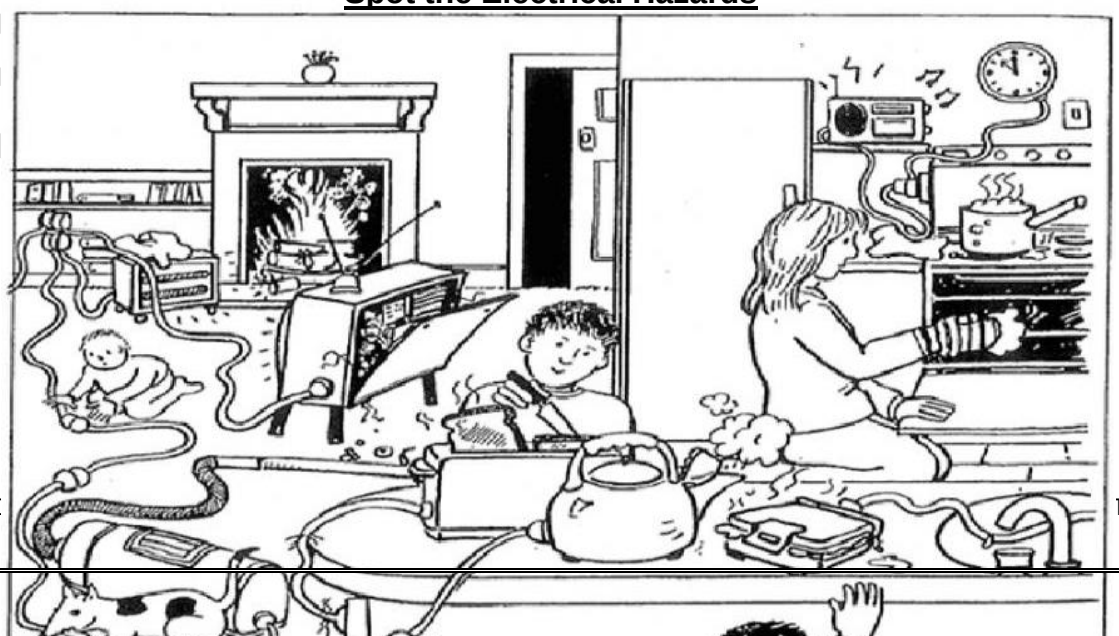
	Clay	Loam	Sand
Size of particles	_____ _____	Loam contains sand and clay particles.	Most of the particles in sandy soil are quite large
Drainage	Clayey soil holds a lot of water	_____ _____	Sandy soil does not hold water well.
Nutrients	Clayey soil has some	Loam contains the right amount of nutrients	_____ _____

3.4 **Draw** an illustration of each soil type below.
Use the correct colour and pay attention to particle size. **3 x ½ = /1½**

Clay	Loam	Sand

Extension Activity
To be completed after you have checked **all your answers** and only if you have finished.

Spot the Electrical Hazards



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